Math 1040

EXAM #3

Name KEY
(print)

Total = 100 points
Please show all your work.

1. (21 pts) Thirty-five percent of adults say almonds are their favorite kind of nut. You randomly select 12 adults and ask each to name his or her favorite nut. Find the probability that the number who will say that almonds are their favorite nut is:

a) (6 pts) exactly 9 $n=12,\ p=.35\ \ q=.65$

$P(9) = .35^9 \times .65^3 \times {}_{12}C_9 = .0048$

Answer .0048 or .48%

b) (9 pts) at most 2

$P(\le 2) = P(0)+P(1)+P(2) = .35^0 \times .65^{12} \times {}_{12}C_0 + .35^1 \times .65^{11} \times {}_{12}C_1 + .35^2 \times .65^{10} \times {}_{12}C_2$
$= .0057 + .0368 + .1088 = .1513$

Answer .1513 or 15.13%

c) (6 pts) Find the mean and the standard deviation of this binominal probability distribution.

$\bar{x} = np = 4.2$ $\sigma = \sqrt{npq} = 1.65$

Mean 4.2 Standard deviation 1.65

2. (10 pts) Find: $P(z < 0.25 \text{ or } z > 2.17)$

$P(z<0.25 \text{ or } z>2.17) = P(z<0.25)+P(z>2.17) = P(z<.25)+1-P(z<2.17)$
$=0.5987 + 1 - .9850 = .6137$

Answer .6137

3. (17 pts) The length of pregnancies of humans are normally distributed with a mean of 268 days and a standard deviation of 15 days.

a) (9 pts) Out of 150 pregnancies, how many would you expect to last more than 265 days?

$\bar{x} = 268 \quad \sigma = 17 \qquad z_{265} = \frac{265-268}{15} = -.2$

$P(z \geq -0.2) = 1 - P(z \leq -0.2) = 1 - .4207 = .5793$

$150 \times .5793 \cong 87$

Answer 87

b) (8 pts) What percentage of pregnancies would last between 255 and 270 days?

$z_{255} = \frac{255-268}{15} = -.87$

$z_{270} = \frac{270-268}{15} = .13$

$P(255 \leq x \leq 270) =$

$P(-.87 \leq z \leq .13) =$

$= P(z \leq .13) - P(z \leq -.87) = .5517 - .1922$

$= .3595$

Answer 35.95% or 36%

4. (20 pts) The number of cats per household is a small town is given. Construct a probability distribution (find P(x)), and then find the mean and standard deviation of the probability distribution.

Cats	Households	$P(x)$	$x-\bar{x}$	$(x-\bar{x})^2$	$(x-\bar{x})^2P(x)$
0	1,270	.605	-.56	.314	.190
1	560	.267	.44	.194	.052
2	208	.099	1.44	2.074	.205
3	52	.025	2.44	5.954	.149
4	10	.005	3.44	11.834	.059
		1.001			.655

$\bar{x} = \sum xP = .56$

$\sigma = \sqrt{.655} = .81$

Mean .56

Standard deviation .81 or .809

5. (15 pts) 1,800 raffle tickets are sold at $5 each for 4 prizes valued at $1,000, $750, $500 and $250. You buy one ticket. What is the expected value of your gain/loss?

$$\frac{995+745+495+245+(-5)\times 1796}{1800} = -3.61$$

Answer -3.61 $

6. (17 pts) The body temperatures of adults are normally distributed with a mean of 98.6° F and a standard deviation of 0.44° F.

a) (9 pts) If you randomly select 50 adults, about how many of them would have the body temperature between 98°F and 99.5°F?

$\bar{x} = 98.6 \quad \sigma = 0.44$

$$z_{98} = \frac{-.6}{.44} = -1.36$$

$$z_{99.5} = \frac{99.5-98.6}{.44} = 2.05$$

$$P(98 \le x \le 99.5) = P(-1.36 \le z \le 2.05) = P(z \le 2.05) - P(z \le -1.36)$$

$$= .9798 - .0869 = .8929$$

$$.8929 \times 50 = 44.65 \simeq 45$$

Answer 45

b) (8 pts) What percentage of adults will have a temperature higher than the value that corresponds to z=1.28?

$$P(z \ge 1.28) = 1 - P(z \le 1.28) = 1 - .8997 = .1003$$

Answer 10 %